RE-MESH

RE-MESH

A self-healing network architecture designed to provide resilient communication in disaster-stricken urban environments.

SLIIT
FACULTY OF COMPUTING

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Mesh Node Hardware & Deployment

Specialization: Computer Systems and Network Engineering

Presenter

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Proven Gap/ Creative Solution

Knowledge Gap

- Existing mesh solutions focus on low-bandwidth text communication, not full IP traffic.
- Lack of field-deployable, low-cost nodes that are rugged, portable, and capable of supporting multiple device types.

Shortcomings of Existing Systems

- Off-the-shelf LoRa/Mesh devices → limited bandwidth and device compatibility.
- Military/enterprise hardware → expensive, heavy, hard to deploy rapidly.

Proposed solution

- ESP32-based portable Wi-Fi mesh nodes with modular power, providing IP connectivity for multiple devices, scalable deployment, and resilient self-healing mesh networking.

Application of Specialization-Based Knowledge

Key Domains Utilized

- Embedded Systems – Designing low-power, modular ESP32-based hardware.
- Wireless Mesh Networking – 802.11s Wi-Fi mesh protocols for self-healing connectivity.
- Power & Energy Management – Integration of battery and solar systems for autonomous operation.

Latest Technologies Applied

- ESP32 microcontroller boards (Wi-Fi + BLE)
- Dual-band antennas for scalable coverage
- Smart battery + solar charge management modules
- Open-source mesh firmware (customizable for field deployment)

Application of Specialization-Based Knowledge

Evaluation / Success Criteria

- Connectivity & Range: Ensure nodes form a stable mesh, covering target areas.
- Throughput & IP Traffic: Support multiple devices, measure network throughput.
- Reliability & Uptime: Battery + solar operation for extended periods (>8 hrs).
- Scalability & Self-Healing: Additional nodes automatically extend mesh.

Data Availability & Ethical Clearance

- Uses open Wi-Fi spectrum → no licensing required.
- No personal or sensitive human data collected → no ethical approval needed.
- Field testing limited to controlled urban and campus environments.

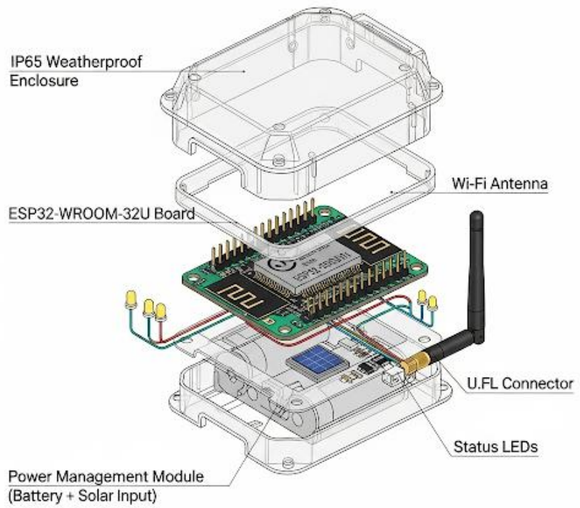
Implementation Details

System Overview

- Node consists of ESP32, battery pack, solar module, antenna
- Configured for plug-and-play field deployment

Real-World Scenario

- Rapid deployment after disaster → drop-in nodes form a resilient mesh
- Survivors and first responders connect via Wi-Fi-enabled devices
- Network expands automatically as more nodes are added



Wireless Communication

Specialization: Computer Systems and Network Engineering

Presenter

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Proven Gap/ Creative Solution

Knowledge Gap:

- Bandwidth vs Range Trade-off
- Deployment Complexity

Addressing Existing System Shortcomings:

- Meshtastic limitations which severely limit disaster response capabilities
- Wi-Fi mesh complexity needs specialized expertise which may unavailable during emergency situations

Proposed Solution:

- Developing wireless peer-to-peer communication protocols for post-disaster emergency response scenarios using dual-protocol mesh architecture.

Application of Specialization Knowledge

Key Domains

- Networking Protocols
- Wireless Technologies
- Embedded Systems

Latest Technologies Used

- ESP-NOW: Connectionless and fast communication
- Wi-Fi 802.11n/g: High-bandwidth client connectivity
- Dual Wi-Fi Mode Support on Microcontrollers: Capable of simultaneous Access Point (AP) and Station (STA) mode
- Adaptive power monitoring using power monitoring sensors (INA3221)

Success Evaluation & Data Availability

- **Validation Metrics:**

Throughput , Latency , Packet delivery rate, Recovery time from node failure

- **Expected Accuracy:**

Network formation , self-healing capability

- **Data Sources:**

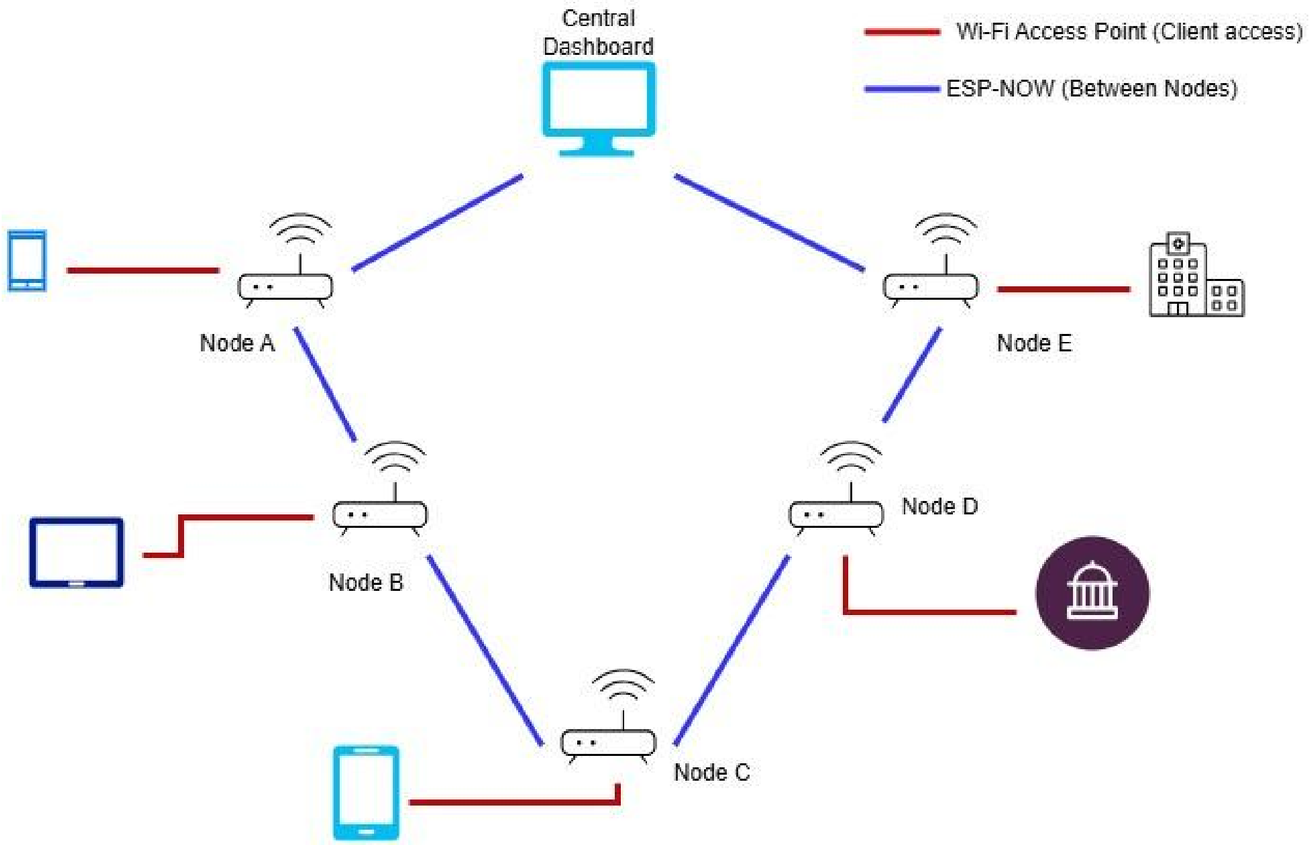
Real-time network performance metrics, power consumption analysis, environmental testing data from urban disaster simulations

Ethical Clearance

- No Personal Data Handling
- Controlled and Field Testing Ethics

Implementation Details

System Architecture Overview



Real-World Use Scenario

- Rapid setup by non-expert emergency personnel
- Nodes automatically discover neighbors
- Maintains communication despite node loss due to power or physical damage

Routing and Data Flow

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Presenter

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Proven Gap & Creative Solution

Proven Gap

- Existing mesh protocols = no emergency priority
- All packets treated equally → delays in disasters

Shortcomings of Existing Systems

- Satellite → costly
- LoRa → long range, very low data rate
- Wi-Fi mesh → good throughput, no prioritization

Proposed Creative Solution

- ESP32-based routing
- Node-level emergency classification
- Priority forwarding of emergencies
- Auto rerouting on node failure

My Scope in RE-MESH

- Design & implement emergency-aware routing logic
- Enhance resilience & responsiveness

Application of Specialization-Based Knowledge

Key Domains:

- Embedded Systems Programming
- Wireless Mesh Networking
- Disaster Communication Protocols

Latest Technologies Used:

- ESP32 microcontrollers with ESP-MESH
- NS-3 network simulator
- Priority Queue Algorithms (custom implementation in C++)

Why These Matter:

- ESP32 offers low-cost, low-power deployment
- NS-3 enables stress testing under simulated disaster conditions
- Priority queues ensure emergency packets are forwarded first

Testing for Real-World Resilience

- **Test Scenarios:**

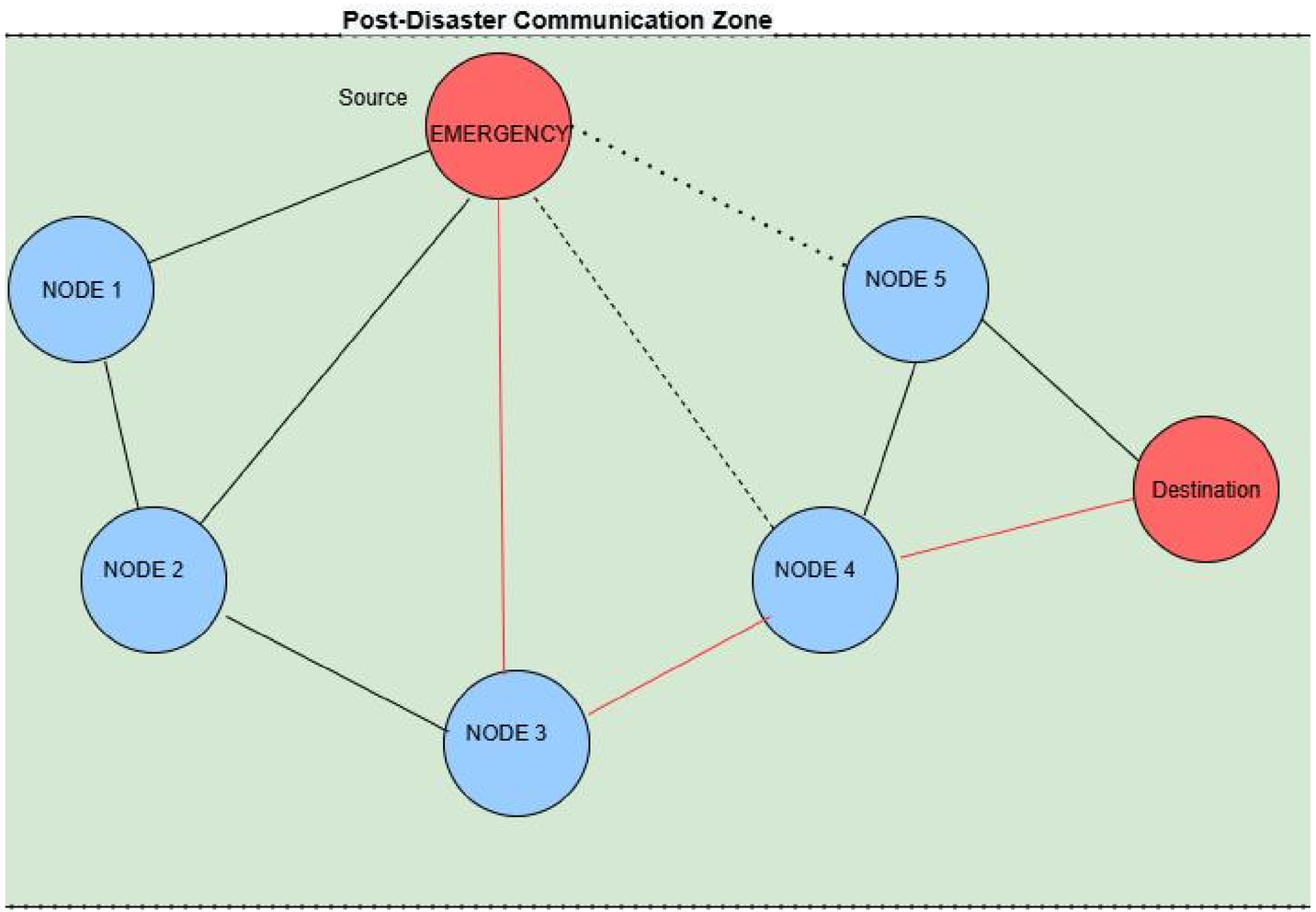
Node failure, congestion, emergency traffic spikes

- **Metrics:**

Latency, packet delivery ratio, hop count

- **Setup:**

4-node ESP32 lab deployment + NS-3 virtual simulation



Implementation & Real-World Use

System Overview:

ESP32 nodes run custom firmware with:

- Packet classification (High/Medium/Low)
- Priority queues
- Adaptive routing logic

Real-World Scenario:

- Deployed in disaster zones with collapsed infrastructure
- Mobile responders carry ESP32 nodes to form instant mesh
- Emergency messages auto-prioritized and rerouted if nodes fail

Centralized Monitoring & Management System

Presenter

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Comparison with Existing Solutions vs Proposed Solution

Proven Gap in Existing Solutions >>>

Satellite Phones, Radio Networks & LoRa-Based Systems:

- No centralized remote configuration
- No built-in device identification mechanism
- Support only voice or text separately
- No real-time visibility.
- Limited scalability, lacking backup & restore for failed nodes.

Proposed Solution >>>

Central Monitoring & Management System for RE-MESH

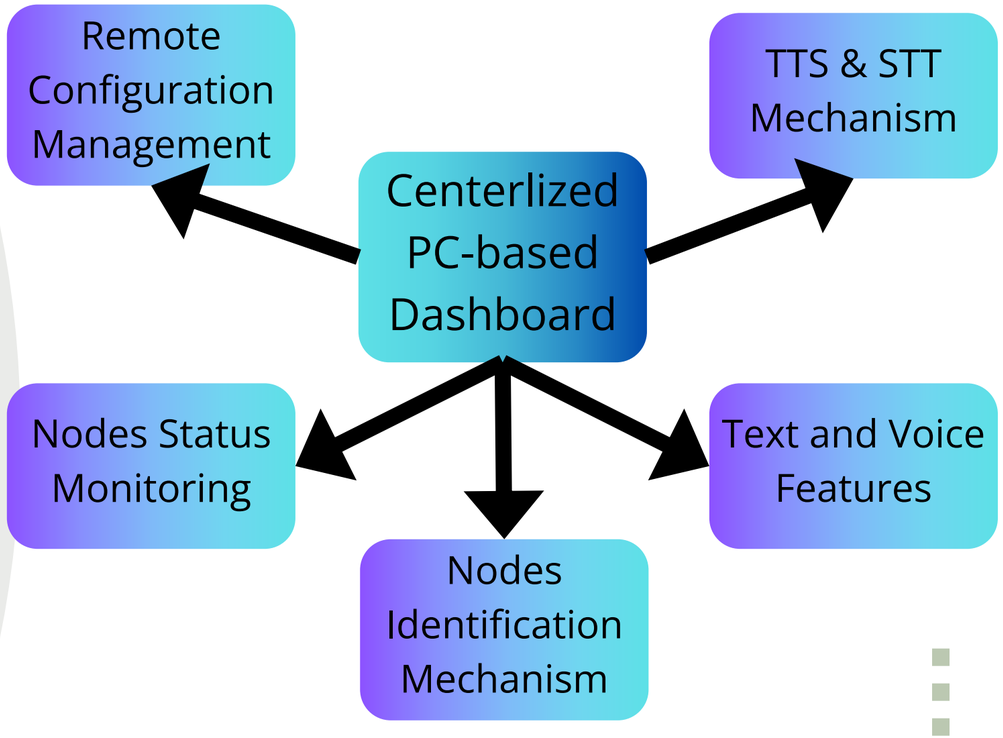
Remote Configuration Management
- Push updates, reconfigure devices, auto-restore failed nodes.

Node Status Monitoring
- Real-time visualization of active/inactive nodes & network topology

Unique Node Identification
- Hardware-based IDs with communication logs for traceability

Integrated TTS & STT
- Converts text → voice and voice → text

Unified Text & Voice Features
- Broadcast and private voice communication.
- Dashboard-to-node text alerts with speech playback.



Application of Specialization-Based Knowledge

Key Domains:

- Systems & Network Engineering – Mesh networking, routing, Wi-Fi.
- Embedded Systems – ESP32 integration with peripherals.
- Cybersecurity – Secure communication.
- Software Engineering – Dashboard and data management.

Latest Technologies Used:

- ESP-WIFI-MESH (self-healing mesh protocol)
- ESP32 Microcontroller with peripherals
- Python GUI Frameworks (Tkinter/PyQt) for dashboard
- TTS & STT Engines (eSpeak, Vosk) for text–voice integration.
- SQLite for offline message & log storage.

Why These Matter:

- Resilient, infrastructure-free communication
- Multi-modal support (text, voice, GPS)
- Scalable & fault-tolerant design
- Cost-effective & accessible solution

Implementation & Real-World Use

System Overview:

- Centralized PC-based Dashboard for monitoring & management.
- Features: Remote configuration, Node status monitoring, Unique ID tracking, TTS/STT integration, and Text/Voice communication.
- Acts as the control center, ensuring resilience and fault tolerance in the mesh network.

How to Use in a Real-World Scenario:

- During a disaster, deployed nodes connect to the dashboard through the mesh
- As the central management system monitor node status, push updates and Ensures continuous communication even if conventional networks fail
- Supports voice broadcasts, unicast and text alerts

Ability of Commercialization

Presenter

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Ability of Commercialization

Investment / Cost & Recovery

- Estimated cost per node: ~LKR 5,000 (material + assembly, max 6–8 units for pilot)
- Investment recovery:
  - Direct sales to NGOs, emergency services, municipal agencies
  - Leasing model for disaster preparedness programs
  - Training & deployment consultancy packages

Target Market & Customer Profile

- **Primary**: NGOs, disaster response agencies, municipal emergency services
- **Secondary**: Community networks, remote rural connectivity projects

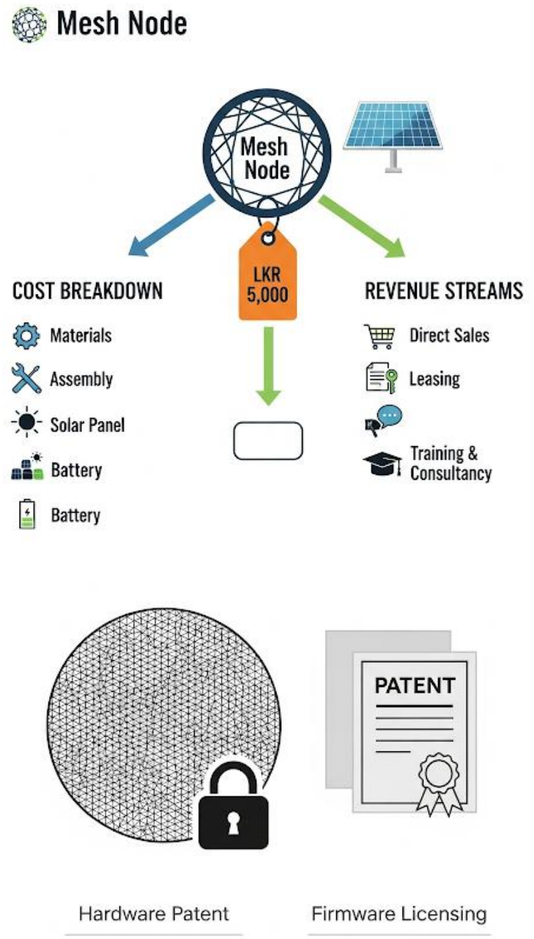
Ability of Commercialization

Value Proposition

- Scalable, self-healing, low-cost, and easy-to-deploy mesh networking solution
- Supports IP traffic for multiple devices, bridging the gap between theory and practical deployment

Intellectual Property & Protection

- Hardware design and deployment methodology patentable
- Open-source firmware can be licensed under commercial-friendly agreements



Q/A
THANK YOU